...ANJALAN SCIENCE AND TECHNOLOGY UNIVERSITY

...of Information and Communication Technology

Class Test No. 03

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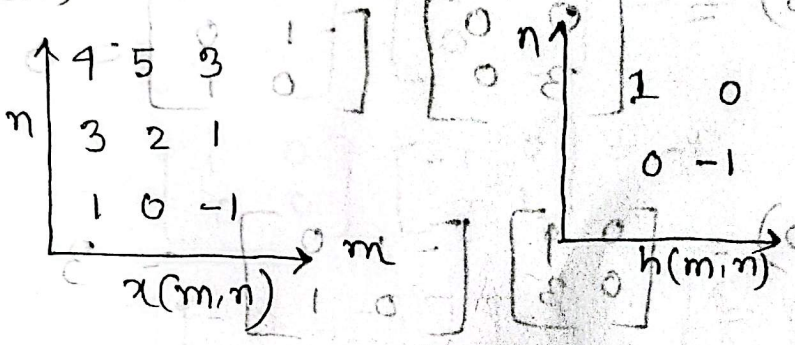
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Course Code ..... Course Title Digital Image Processing

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Ans. to the question no-04

Given,



Flipped kernel is, $h_f = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$, output size = 4×

After padding with zeroes:

and $y_p = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 4 & 5 & 3 & 0 \\ 0 & 3 & 2 & 1 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

P.T.O.

$$y(0,0) = \begin{bmatrix} 0 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = 4$$

$$y(0,1) = \begin{bmatrix} 0 & 0 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = 5$$

$$y(0,2) = \begin{bmatrix} 0 & 0 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = 3$$

$$y(0,3) = \begin{bmatrix} 0 & 0 \\ 3 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = 0$$

$$y(1,0) = \begin{bmatrix} 0 & 4 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = 3$$

$$y(1,1) = \begin{bmatrix} 4 & 5 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = -4+2 = -2$$

$$y(1,2) = \begin{bmatrix} 5 & 3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = -5+1 = -4$$

$$y(1,3) = \begin{bmatrix} 3 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = -3$$

$$y(2,0) = \begin{bmatrix} 0 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = 1$$

$$y(2,1) = \begin{bmatrix} 3 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = -3$$

CamScanner

$$y(2,2) = \begin{bmatrix} 2 & 1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = -2 - 1 = -3$$

$$y(2,3) = \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = -1$$

$$y(3,0) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = 0$$

$$y(3,1) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = -1$$

$$y(3,2) = \begin{bmatrix} 0 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = 0$$

$$y(3,3) = \begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = 1$$

$$\therefore y(m,n) = \left[\begin{array}{cccc} 4 & 5 & 3 & 0 \\ 3 & -2 & -4 & -3 \\ 1 & -3 & -3 & -1 \\ 0 & -1 & 0 & 1 \end{array}\right]$$

Ans. to the question no-03

The conditions of histogram equalization:

i) $T(r)$ should be monotonic.

ii) $0 \leq T(r) \leq L-1$

Ans. to the question no-02

Histogram: A histogram is a graphical representation of the distribution of pixel intensity values in an image.

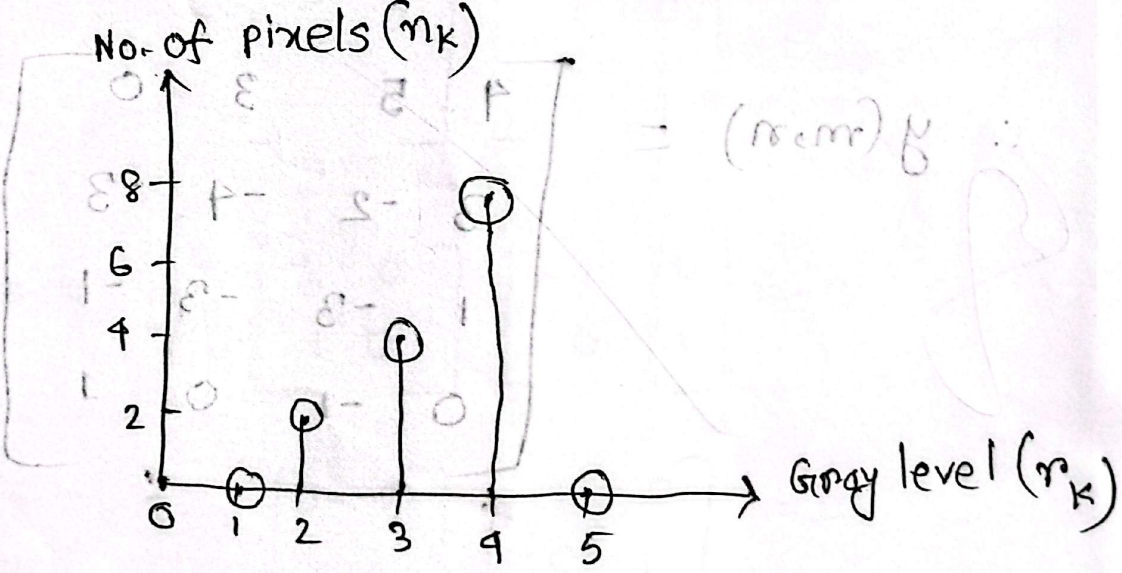


Fig: histogram of an image.

When we will use local histogram enhancement technique:

local histogram enhancement technique is used to improve the contrast of small regions of intensity values. It enhances local details and extracts features that may not be visible by global histogram equalization specially in image with non-uniform illumination.

Applications:

- Medical Image processing.
- Fingerprint recognition
- Surveillance systems
- low light enhancement systems.
- Pattern recognition.

Ans. to the question no-01

Convolution : Convolution is a mathematical operation used in digital image processing to apply a kernel or mask to an image. It is performed by rotating the kernel by 180° and then computing the weighted sum of the neighbouring pixels.

$$g(x,y) = f(x,y) * h(x,y)$$

or,

$$f(x) * g(x) = \int_{-\infty}^{\infty} f(\alpha)\, g(x-\alpha)\, d\alpha$$

Correlation: Correlation is a mathematical operation that is used in digital image processing to measure the similarity between an image and template.

$$g(x,y) = f(i,j)\, h(i,j)$$

or,

$$f(x) * g(x) = \int_{-\infty}^{\infty} f(\alpha)\, g(x+\alpha)\, d\alpha$$